

50.051 Programming Language Concepts

Course Introduction

Matthieu De Mari

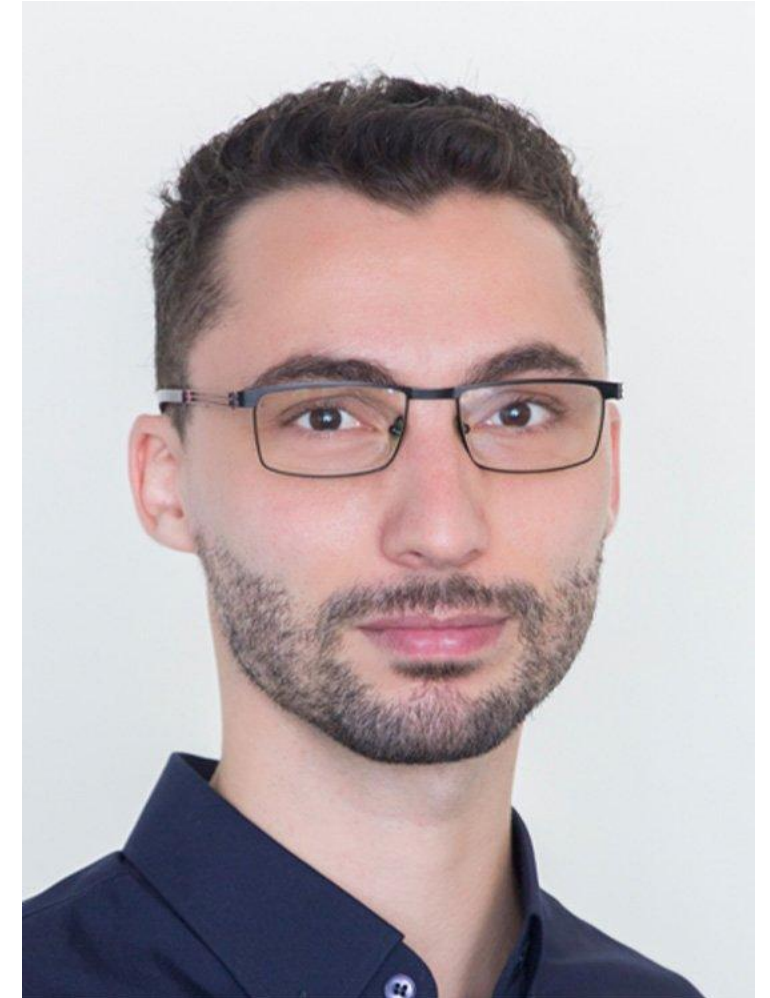


SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

A quick word about instructors

Matthieu (Matt) De Mari

- Senior Lecturer at SUTD (Python, Deep Learning, Prog. Language Concepts, Capstone, etc.)
- Director of the Master of Science in Design and Artificial Intelligence for Enterprise (MSc DAI-E)
- PhD from CentraleSupélec (France)
- Email: matthieu_demari@sutd.edu.sg
- Office @ SUTD: 2.401-07 (located in SUTD Academy/AI cluster).



A quick word about instructors

Dileepa Fernando

- Lecturer at SUTD (Python, Software Engineering, Programming Language Concepts and more)
- Computer Science and Design (CSD)
- PhD from NUS
- Email: dileepa_fernando@sutd.edu.sg
- Office @ SUTD: 1.402.24



A quick word about instructors

Juan David Salazar Rodríguez (TA)

- PhD candidate in the ASD pillar at the Singapore University of Technology and Design (SUTD)
- Research on artificial intelligence as a collaborative agent in architectural education, focusing on LLMs, AI personas, and mixed reality systems to support design critique, reflection, and creative learning in studio-based pedagogy.
- Contact: 1008372@mymail.sutd.edu.sg



Course Handout/Syllabus

- Everything you need to know about the course is in the course handout on eDimension.
- Might get updated with more details (dates, locations, etc.) about exams when OSA confirms this with us.
- **Before you ask any questions, check the course handout.**

Course structure

Part 1: C programming + State Machines (6 weeks)

- Everything about C
- Memory Management
- Finite State Machines and Regular Expressions

Part 2: Compiler Engineering & C++ (6 weeks)

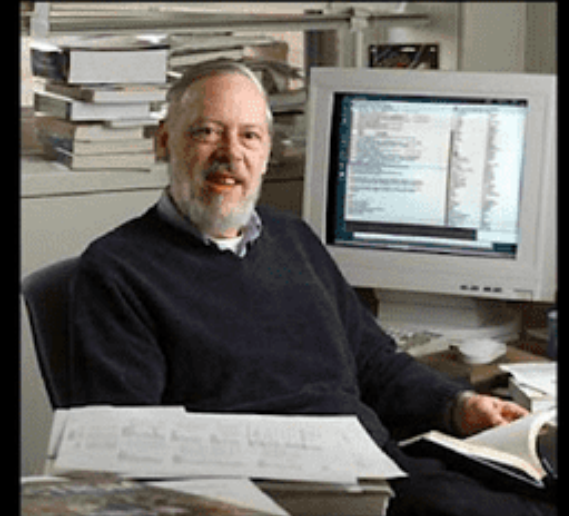
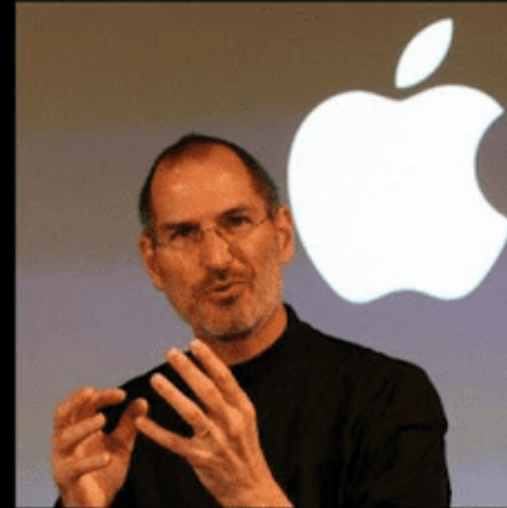
- Lexical/Syntactic/Semantic Analysis
- Intermediate Code Representations, Optimization and Backend
- C++

About C

C created in 1972 by Dennis Ritchie between 1969-1973

- Called C as a successor to B
- Imperative programming language
 - Program = set of statements

C inspired many other languages and is a good entry point for learning many additional languages (C++, Objective-C, Java, C#, PHP, etc.)



Your attention, please:

Both of these men died in the same month of the same year. Steve was largely considered a hero, while Dennis was largely ignored by the world. Only a handful of programmers who know the real value of Dennis Ritchie's work even know of his death.

Without Steve Jobs there is no iPhone, iPad, iPod or Macintosh. Without Dennis there is no C. Without C, there is no Unix, Windows, or Linux. Without C there is no C++ nor Objective C. There is no MacOS X, no iOS, no Photoshop, no FLStudio, no Firefox, no Safari, no Google Chrome, no Playstation, no Xbox. In fact, 90% of the applications in the world are written in C or C++ or Objective C. **If you think Dennis deserves our respect, please pass this along.**

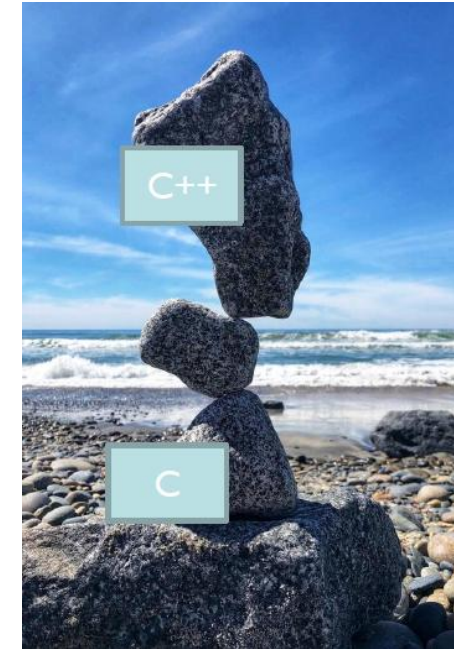
Why should I learn C?

- C is still widely used for operating systems (Unix, Linux, Windows, Mac OS X, etc.)
- Can be compiled on nearly any hardware
- “Low level language”
 - Great for memory management
 - Small set of keywords
- Still among the most popular and used languages, even in 2025.



Why should I learn C++?

- Created in 1985
- C++ = "C with Classes"
- Kinda like Java
 - More "powerful"?
 - More complicated
- If you know Java and C++, you will have no trouble understanding C#.
- These days, C++ used in many AI frameworks (faster than Python for training/inference)



PyTorch

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Shortcuts

NOTE: Latest Stable PyTorch requires Python 3.10 or later.

Prerequisites		PyTorch Build	
Supported Windows		Linux	Mac
Distributions		Pip	LibTorch
Python		Language	C++ / Java
Package Manager		Compute Platform	CUDA 12.6
Installation			CUDA 12.8
pip			CUDA 13.0
Verification			ROCm 6.4
Building from source			CPU
Prerequisites		Run this Command:	

Download here (Release version):
<https://download.pytorch.org/libtorch/cu126/libtorch-win-shared-with-deps-2.9.1%2Bcu126.zip>

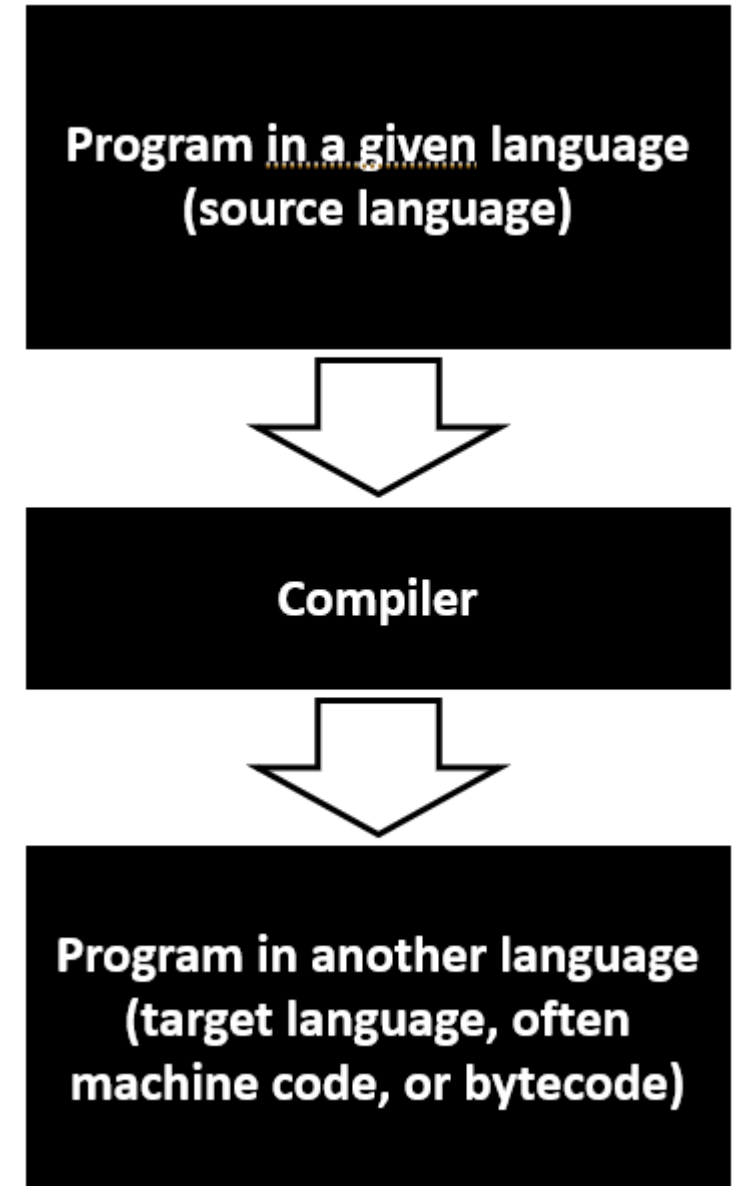
Download here (Debug version):
<https://download.pytorch.org/libtorch/cu126/libtorch-win-shared-with-deps-debug-2.9.1%2Bcu126.zip>

What is compilation?

Compilation = How to go from source code in any programming language to assembly

- Tokenization
- Lexical Analysis
- Syntactic Analysis
- Semantic Analysis
- Parsing
- And more!

Compilers = Missing link between your beloved programming languages and 50.002 CompStruct



Why should you study compilers in a Computer Science degree?

1. A great introduction to techniques for translation and improvement of programs written in any programming language.
2. A fantastic practical exercise in software engineering.
3. Provides an understanding of what happens in the background when you press the “Run” button of your programs.
4. A way to understand where typical compilation errors come from.
5. A better understanding of the missing link between any programming course and the 50.002 Computation Structures one.
6. A much-needed introduction course to compilers, if one day you plan on writing your own programming language (*please don't!*).

Final Grade

- **Practicals (20%):** 6 individual assignments
 - Given in class, either check off in class or due 2 weeks after session
- **Group Project (20%):** 4 students per group max
 - Instructions given in week 2-3
- **Class participation + Student Feedback (8+2%)**
- **Midterm Exam (20%)**
- **Final Exam (30%)**

All details in the course handout.